CHALUMEAU FAMILY CAPSTONE PRINT PACKET

Generated: 2026-05-03
Packet folder: /mnt/c/Users/Tony/Documents/GitHub/chalumeau

FILE MAP

File	Purpose
design.md	Project intent, catalog metadata, assumptions, and validation plan.
bom.csv	Starter bill of materials with part categories, quantities, drawing refs, and notes.
sourcing.csv	Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.
cut-list.csv	Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.
drawing-brief.md	Manufacturing drawing and technical product sketch brief.
assembly-manual.md	Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.
validation.csv	Target/measured values, tolerance, environment, result, and tuning/build action log.
supplier-rfq.md	Supplier email/request-for-quote starter.
visual-bom-brief.md	Art direction for an image-forward visual BOM.
wolfram-starter.wl	Wolfram starter for physics, optimization, visualization, and validation.
README.md	Project artifact.
SKILLS.md	Project artifact.
family-spec.csv	Project artifact.

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DESIGN.MD

Project intent, catalog metadata, assumptions, and validation plan.

CLM-001 - CHALUMEAU FAMILY

Generated: 2026-05-03

INTENT

Design a family of chalumeaux that can be built in Tony’s shop using lathe/CNC workflows, commercial reeds for early isolation tests, and optional handmade metal keywork. The family should cover a useful range without pretending the first-pass physics is final: all tone-hole coordinates are starting points that must be validated with a real reed, mouthpiece, bore, and player pressure.

SOURCE ARTIFACTS

- Repo: https://github.com/tonykoop/chalumeau
- Local inspiration photos: images/chalumeau1.jpg, images/chalumeau5.jpg, and images/7173-372-1_1920x1080.avif
- Workbook inspected: C:/Users/Tony/Documents/Claude/Projects/Career/flutes-staging/Musical Instruments.xlsx
- Relevant workbook source sheet: Great Highland Bagpipe, especially the contrast between conical double-reed chanter rows and cylindrical single-reed drone rows.
- Done-bar reference style: tongue-drum repo packet and skill v3 packet requirements.

GOVERNING MODEL

The chalumeau is modeled as a cylindrical single-reed pipe that behaves like a stopped pipe in its low register.

text

```
f = c / (4  L_eff)
L_eff = c / (4  f)
c = 13552 in/s at about 68 F
body_length = L_eff - bell_end_correction - reed_end_correction
bell_end_correction = 0.61  bore_radius
reed_end_correction = 0.25  bore_id   # first-pass assumption
tone_hole_x_from_bell = body_length - (c/(4f_note) - reed_corr - 0.30hole_dia)
cents_error = 1200  log2(measured_hz / target_hz)
```

The model deliberately does not use Tony’s NAF K2 corrections. Those corrections are for open-open Native American style flutes in a specific bore range. The chalumeau reed, stopped-pipe boundary condition, mouthpiece volume, bore diameter, and tone-hole chimneys need their own empirical correction after prototype measurement.

FAMILY SPEC

ID	Variant	Root	Bore ID	Final body L	Blank L	Sections
CLM-SOP-C4	Soprano C	C4	0.5	12.672	14.172	1 body + mouthpiece + bell
CLM-ALT-G3	Alto G	G3	0.625	16.939	18.439	1 body + mouthpiece + bell
CLM-TEN-C3	Tenor C	C3	0.75	25.483	26.983	2 body joints + mouthpiece + bell
CLM-BAS-F2	Bass F	F2	0.875	38.32	39.82	3 body joints + mouthpiece + bell

TONE-HOLE STRATEGY

- The keyless core uses seven front holes for a diatonic octave: offsets +2, +4, +5, +7, +9, +11, +12 semitones from the all-closed root.
- K01 is an optional normally closed semitone key near the bell. It gives the first chromatic note above the root without forcing an awkward low finger stretch.
- K02 is an optional upper chromatic side key. It gives a practical chromatic note between scale degrees 6 and 7 and doubles as a keywork fabrication exercise.
- A clarinet-style register vent is listed only as an experimental future feature. The first chalumeau should be judged by the low register before chasing overblown twelfths.

HARDWARE ALIGNMENT

The hardware is intentionally small-shop manufacturable:

Hardware	Role	First-build method	Upgrade path
K01 lever	opens low semitone tone hole	brass strip lever, pivot post pair, leather/cork pad	nickel-silver lever with soldered pad cup
K02 lever	opens upper chromatic tone hole	side lever with flat spring return	clarinet-style post/rod key with regulation cork
Optional register vent	tests clarinet evolution	leave undrilled until the body speaks well	small lined vent tube near mouthpiece
Raised tone-hole collars	ergonomic/aesthetic reference to photos	turn integral collars or add rings	separate stabilized-wood collars

SOLIDWORKS PARAMETER CONVENTION

Use CLM_ as the project prefix, keep units in the variable name, and use stable hole IDs that match the CSV and drawings:

- CLM_Bore_ID_in, CLM_Body_L_Final_in, CLM_Bell_OD_in
- CLM_H01_X_Bell_in, CLM_H01_Dia_in
- CLM_K01_X_Bell_in, CLM_K01_Dia_in
- CLM_Key_Pivot_Rod_Dia_in, CLM_Pad_Overhang_in

The SolidWorks design table is cad/solidworks-design-table.csv. The equation snippets are in cad/solidworks-equations.txt.

ASSUMPTIONS AND RISKS

Risk	Why it matters	Mitigation
Reed/mouthpiece compliance shifts pitch	the reed is not a hard closed end	test with

one commercial reed/mouthpiece before final hole tuning |
Tone holes are first-pass estimates	small holes do not act as perfect open ends	drill undersized, tune larger, log every change
Bass holes exceed finger comfort	low instruments become large quickly	use key pads for lower/bigger holes
Handmade key pads leak	leaks ruin tuning and response	leak-light/suction test before pitch tuning
Dense oily woods complicate machining	cocobolo/ipe dust and finish issues	prototype in cherry/maple first

VALIDATION PLAN

1. Calibrate tuner/microphone at A4 = 440 Hz.
2. Build CLM-SOP-C4 with no levers and a commercial reed/mouthpiece.
3. Tune the all-closed root by trimming the bell/foot only after reed response is stable.
4. Open holes from the bell upward, enlarging in small increments and logging measured frequency and cents error.
5. Add K01/K02 to either the same body or a second soprano body. Validate sealing before measuring pitch.
6. Update the empirical correction column in the design workbook before scaling to alto, tenor, or bass.

NEXT ACTIONS

- Confirm the actual reed and mouthpiece family for the soprano prototype.
- Decide whether the first body gets raised collars like the inspiration image or plain chamfered holes for faster tuning.
- Model CLM-SOP-C4 in SolidWorks using the provided global variables.
- After first measurements, add a prototype_correction_pct column to data/tone-hole-schedule.csv.

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BOM.CSV

Starter bill of materials with part categories, quantities, drawing refs, and notes.

item_no	subsystem	part_name	qty	material_spec	dimensions_spec	make_buy	estimated_cost	drawing_ref	notes
1	Body	Turned chalumeau body blanks	4	Cherry or hard maple for first pass; ipe/cocobolo/boxwood for keeper builds	see family-spec.csv blank_length_in and turning_blank_in	make	80	drawings/chalumeau-family-sheet.svg	Use lower-cost domestic hardwood for bore and keywork tests before dense/oily woods.
2	Bore	Long drill and reamer set	1	Aircraft-length brad point/twist drills plus adjustable/shell reamers	0.500, 0.625, 0.750, 0.875 in bore targets	buy	120	cnc/cnc-lathe-plan.md	Exact tooling depends on whether bores are drilled solid or routed as

split blanks. |
3	Mouthpiece	Single-reed mouthpieces	4	Purchased clarinet/sax mouthpiece adapted by size, or shop-made Delrin mouthpiece	socket IDs in family-spec.csv; reed table in design.md	buy first, make later	160	drawing-brief.md	Use bought mouthpieces/reeds to isolate body acoustics before making reeds.
4	Tone holes	Raised tone-hole collars or drilled holes	36	Integral turned collars for visual style; chamfered holes for first prototype	see data/tone-hole-schedule.csv	make	20	drawings/chalumeau-soprano-c4-dimensioned.svg	Collars are visual and ergonomic; pitch comes from hole center, diameter, and chimney.
5	Keywork	Two-key chalumeau lever set	4	0.040-0.062 in brass/nickel silver sheet, 1/16 in pivot rod, phosphor bronze spring	K01 and K02 per variant; see hardware/keywork-parts.csv	make	60	drawings/keywork-lever-detail.svg	The first soprano can be built keyless, then retrofitted with keys.
6	Pads	Leather/cork/felt pad stack	12	Clarinet-style skin pads or cork backed with thin leather	pad OD = tone hole OD + 0.080 to 0.125 in	buy or make	35	hardware/lever-fabrication-guide.md	Use leak light or suction test before tuning.
7	Finish	Oil/shellac finish and bore oil	1	Dewaxed shellac outside; bore oil compatible with chosen wood	finish schedule in assembly-manual.md	buy	30	assembly-manual.md	Avoid thick finish buildup inside tone holes or under pads.

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SOURCING.CSV

Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.

component	required_spec	search_terms	supplier_candidates	date_checked	unit_price	lead_time	substitute_rule	risk_notes
Domestic hardwood prototype blanks	Straight-grain cherry, hard maple, or walnut; 1.5-2.25 in square; length per family-spec.csv	turning blank cherry hard maple 2x2x18 2x2x30	local hardwood dealer, Bell Forest, Cook Woods, Woodcraft	not live-checked	research before purchase	research before purchase	Any stable fine-grained hardwood is acceptable for first acoustic prototypes.	Soft woods dent around tone holes and key posts; use for process tests only.
Dense keeper-build blanks	Ipe, cocobolo, boxwood, rosewood, grenadilla substitute; dry and crack-free	ipe turning blank cocobolo clarinet blank boxwood woodwind blank	Gilmer Wood, Cook Woods, Bell Forest, specialty woodwind blank suppliers	not live-checked	research before purchase	research before purchase	Delrin/acetal is acceptable for dimensional-stability testing.	Allergenic/oily woods need dust control, wiping before glue, and finish tests.
Single reeds and mouthpieces	Soprano/alto use clarinet-like reeds; tenor/bass may use sax/low clarinet reeds or custom mouthpiece	clarinet mouthpiece blank baroque clarinet reed tenor chalumeau mouthpiece	woodwind suppliers, early-music makers, 3D print/Delrin shop-made option	not live-checked	research before purchase	research before purchase	Use commercial reed first to decouple reed-making from bore tuning.	Reed strength and mouthpiece volume can shift pitch and response significantly.
Brass/nickel-silver key stock	0.040-0.062 in sheet, 1/16 in rod, 0-80 or M1.6 screws,

post stock | nickel silver sheet woodwind key rod phosphor bronze spring wire | McMaster-Carr, K&S metals, MusicMedic, instrument repair suppliers | not live-checked | research before purchase | research before purchase | Brass is easier to prototype; nickel silver wears better and solders cleanly. | Tiny screws strip easily; buy extras and make drill guides. |
| Pads, cork, felt, leather | Clarinet/sax pads or leather-over-cork pads sized to K01/K02 tone holes | clarinet pads sheet cork felt key bumper leather pad cup | MusicMedic, Ferree’s Tools, instrument repair suppliers | not live-checked | research before purchase | research before purchase | Cork plus thin leather is acceptable for first handmade keys. | Pad seating matters more than pad material for early prototypes. |

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CUT-LIST.CSV

Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.

part_id	variant	material	qty	rough_dimensions_in	final_dimensions_in	operation	grain_orientation	yield_notes
CLM-SOP-C4-BODY	Soprano C	prototype hardwood	1	1.25 x 1.25 x 14.172	OD 0.875; bore 0.5; body length 12.672	lathe turn, drill/ream bore, drill tone holes, final tune	long grain parallel to bore	leave 1.5 in length allowance and 0.1875 in OD cleanup allowance
CLM-SOP-C4-BELL	Soprano C	contrasting hardwood or same blank offcut	1	2.025 x 2.025 x 3.000	bell OD 1.65; socket fits bore/body OD by CAD	lathe turn bell flare and socket	long grain parallel to bore	use a separate bell to simplify tuning and repair
CLM-ALT-G3-BODY	Alto G	prototype hardwood	1	1.438 x 1.438 x 18.439	OD 1.063; bore 0.625; body length 16.939	lathe turn, drill/ream bore, drill tone holes, final tune	long grain parallel to bore	leave 1.5 in length allowance and 0.1875 in OD cleanup allowance
CLM-ALT-G3-BELL	Alto G	contrasting hardwood or same blank offcut	1	2.325 x 2.325 x 3.000	bell OD 1.95; socket fits bore/body OD by CAD	lathe turn bell flare and socket	long grain parallel to bore	use a separate bell to simplify tuning and repair
CLM-TEN-C3-BODY	Tenor C	prototype hardwood	1	1.625 x 1.625 x 26.983	OD 1.25; bore 0.75; body length 25.483	lathe turn, drill/ream bore, drill tone holes, final tune	long grain parallel to bore	leave 1.5 in length allowance and 0.1875 in OD cleanup allowance
CLM-TEN-C3-BELL	Tenor C	contrasting hardwood or same blank offcut	1	2.725 x 2.725 x 3.000	bell OD 2.35; socket fits bore/body OD by CAD	lathe turn bell flare and socket	long grain parallel to bore	use a separate bell to simplify tuning and repair
CLM-BAS-F2-BODY	Bass F	prototype hardwood	1	1.875 x 1.875 x 39.82	OD 1.5; bore 0.875; body length 38.32	lathe turn, drill/ream bore, drill tone holes, final tune	long grain parallel to bore	leave 1.5 in length allowance and 0.1875 in OD cleanup allowance
CLM-BAS-F2-BELL	Bass F	contrasting hardwood or same blank offcut	1	3.275 x 3.275 x 3.000	bell OD 2.9; socket fits bore/body OD by CAD	lathe turn bell flare and socket	long grain parallel to bore	use a separate bell to simplify tuning and repair

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DRAWING-BRIEF.MD

Manufacturing drawing and technical product sketch brief.

CHALUMEAU DRAWING BRIEF

Instrument: Chalumeau family
Revision/date: REV-A
Units: inches unless noted
Source workbook/CAD/catalog ID: CLM-001 / SolidWorks design table

REQUIRED VIEWS

- Family comparison side view with bore axis and body lengths.
- Soprano C dimensioned side view with all H/K hole coordinates.
- Cross-section through bore, tone-hole chimney, and raised collar.
- Keywork detail for K01/K02: closed/open positions, pad, cup, pivot, posts, spring, touchpiece.
- Exploded view: mouthpiece/reed, body, bell, K01, K02, pads, posts.
- Ergonomic view: hand reach for keyless soprano and keyed bass notes.

CRITICAL DIMENSIONS

Feature	Dimension source	Tolerance
Bore ID	family-spec.csv	+/- 0.003 in after ream/lap
Body length	family-spec.csv	leave trim allowance; final by tuning
Hole X from bell	data/tone-hole-schedule.csv	+/- 0.020 in first prototype
Hole diameter	data/tone-hole-schedule.csv	start undersize; final by tuning
Pad seat flatness	hardware/keywork-parts.csv	no visible leak
Pivot alignment	keywork drawing	lever moves freely without side shake

DRAWING OUTPUTS IN THIS PACKET

- drawings/chalumeau-family-sheet.svg
- drawings/chalumeau-soprano-c4-dimensioned.svg
- drawings/keywork-lever-detail.svg

NOTES FOR SOLIDWORKS

Drive dimensions from global variables. Do not dimension hole centers manually in separate sketches unless the dimension names match the CSV.

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ASSEMBLY-MANUAL.MD

Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.

CHALUMEAU ASSEMBLY MANUAL

BUILD ORDER

Build the soprano C keyless version first. Do not add keywork until the bore, reed, and seven front holes speak reliably.

PHASE 0 - DESIGN FREEZE

1. Pick the exact reed and mouthpiece.
2. Print data/tone-hole-schedule.csv and mark which holes will be drilled in the first pass.
3. Decide whether the prototype uses raised collars or plain chamfered holes. Plain holes tune faster; collars match the inspiration photo.
4. Confirm the blank dimensions from cut-list.csv.

PHASE 1 - BODY BLANK AND BORE

1. Mill/turn the blank oversize and mark the bore centerline.
2. Drill the bore from the reed-seat end with the longest stable drill setup available.
3. Ream or lap to final bore diameter. Measure at both ends.
4. Turn the exterior body profile, leaving extra length at the bell end.
5. Cut the mouthpiece socket/tenon but keep the reed setup removable.

PHASE 2 - ROOT TUNING

1. Assemble reed, mouthpiece, and body with all holes undrilled.
2. Measure the all-closed pitch.
3. Trim the bell/foot in small steps only after the reed is stable.
4. Record every trim amount in validation.csv.

PHASE 3 - TONE HOLES

1. Lay out hole centers from data/tone-hole-schedule.csv, measuring from DATUM_BELL_FACE.
2. Center punch lightly. A wandering bit will change tuning.
3. Drill each hole 15-25 percent undersize.
4. Open holes from low to high. After each enlargement, play the target note and log measured frequency.
5. Chamfer only after pitch is close; chamfering can sharpen the note.

PHASE 4 - OPTIONAL K01/K02 KEYWORK

1. Drill keyed tone holes undersize.
2. Fabricate levers using hardware/lever-fabrication-guide.md.
3. Install posts and pivot rods.
4. Seat pads and leak-test before tuning.
5. Tune keyed notes after the pad seal is stable.

PHASE 5 - FINISH

1. Sand exterior through 320 or finer.
2. Seal the bore lightly; avoid thick buildup.
3. Apply shellac/oil finish outside.
4. Keep finish out of tone holes and under pads.
5. Re-test all notes after finish cures.

MAINTENANCE NOTES

- Swab the bore after playing.
- Check pad seating after humidity swings.
- Keep a build log for each reed/mouthpiece combination; reed strength can make the same body read sharp or flat.

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VALIDATION.CSV

Target/measured values, tolerance, environment, result, and tuning/build action log.

test_id	variant	target	target_value	measured_value	tolerance	environment	result	action
VAL-001	all	A4 reference	440.00 Hz		tuner reads 440 within calibration	record temp F, humidity percent		Calibrate tuner/mic before instrument data.
VAL-010	Soprano C	all holes closed	C4, 261.63 Hz		+/- 15 cents before final tuning; +/- 8 cents after	68-72 F preferred		Trim bell end shorter to sharpen; add temporary tape/extension if sharp.
VAL-020	all	tone-hole sweep	each row in data/tone-hole-schedule.csv		+/- 12 cents after tuning	same reed, same mouthpiece, same blowing pressure		Open hole diameter to sharpen; use wax/tape collar to flatten during tests.
VAL-030	keyed builds	pad leak test	no visible leak under leak light; suction holds 10 s		zero audible hiss at normal blowing pressure	before final tone-hole tuning		Reseat pad, regulate cork, or lap tone-hole rim.
VAL-040	experimental register key	overblown twelfth check	note + 19 semitones if vent is clarinet-like		document response only; not a first-build pass/fail	same reed and embouchure as low-register test		Move/resize vent only after the core chalumeau speaks well.

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SUPPLIER-RFQ.MD

Supplier email/request-for-quote starter.

SUPPLIER RFQ - CHALUMEAU MATERIALS AND KEYWORK

Hello,

I am sourcing materials for a small family of handmade single-reed chalumeaux. Please quote the items below, including unit price, volume price if applicable, lead time, shipping estimate, and any recommended substitutes.

WOOD OR PLASTIC BLANKS

- Straight-grain turning blanks suitable for woodwind bodies.
- Prototype species: cherry, hard maple, walnut, or Delrin/acetal.
- Keeper species: ipe, boxwood, cocobolo, rosewood, or grenadilla substitute.
- Sizes range from about 1.25 x 1.25 x 15 in to 2.25 x 2.25 x 41 in. Exact dimensions are in cut-list.csv.

KEYWORK MATERIALS

- Brass or nickel-silver sheet, 0.040-0.062 in thick.
- 1/16 in rod or equivalent pivot stock.
- Small screws, post stock, and phosphor bronze spring wire.
- Clarinet/sax pads or pad leather/cork materials.

REQUIREMENTS

- Material must be stable, dry, and suitable for fine drilling/turning.
- Please identify any allergy/dust or finish compatibility concerns.
- Substitutions are acceptable if dimensional stability and machinability are comparable.

Thank you.

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VISUAL-BOM-BRIEF.MD

Art direction for an image-forward visual BOM.

VISUAL BOM BRIEF

LAYOUT

Use the Ashiko visual BOM pattern: hero image at the top, spreadsheet rows below, and one part image/render per row. Use the actual local chalumeau photos for the hero/inspiration image until Tony has shop photos of the prototypes.

REAL IMAGES AVAILABLE

- images/chalumeau1.jpg: full instrument inspiration.
- images/chalumeau5.jpg: close-up of raised collars and wood finish.
- images/7173-372-1_1920x1080.avif: additional inspiration image, not universally supported by all renderers.

PART ROWS TO SHOW

1. Turned body blank.
2. Mouthpiece/reed assembly.
3. Bell.
4. H01-H07 tone-hole row with raised collars.
5. K01 low semitone lever.
6. K02 side chromatic lever.
7. Pad/cork/felt stack.
8. Pivot posts, rods, springs, screws.
9. Finish and bore oil.

PLACEHOLDER POLICY

Generated or schematic part images are acceptable for planning, but mark them as placeholders until replaced with supplier images or shop photos. Build-critical dimensions must come from the CSV/workbook/SolidWorks, not from pixels in the visual BOM.

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WOLFRAM-STARTER.WL

Wolfram starter for physics, optimization, visualization, and validation.

wolfram

```
( Chalumeau family acoustic starter - generated 2026-05-03 )

ClearAll["Global"];

speedOfSoundInPerSec = 13552.0;
midiFrequency[m_] := 4402^((m - 69)/12);
centsError[measured_, target_] := 1200Log2[measured/target];

bodyLength[rootMidi_, boreID_] := Module[
{f = midiFrequency[rootMidi], acoustic, bellCorr, reedCorr},
```

```
acoustic = speedOfSoundInPerSec/(4f);
bellCorr = 0.61(boreID/2);
reedCorr = 0.25boreID;
acoustic - bellCorr - reedCorr
];

holeXFromBell[rootMidi_, offset_, boreID_, holeRatio_] := Module[
{f = midiFrequency[rootMidi + offset], holeDia, holeCorr, reedCorr, eff, body},
holeDia = boreIDholeRatio;
holeCorr = 0.30holeDia;
reedCorr = 0.25boreID;
eff = speedOfSoundInPerSec/(4f);
body = bodyLength[rootMidi, boreID];
body - (eff - reedCorr - holeCorr)
];

variants = {
< | "ID" -> "CLM-SOP-C4", "RootMIDI" -> 60, "BoreID" -> 0.500 | >,
< | "ID" -> "CLM-ALT-G3", "RootMIDI" -> 55, "BoreID" -> 0.625 | >,
< | "ID" -> "CLM-TEN-C3", "RootMIDI" -> 48, "BoreID" -> 0.750 | >,
< | "ID" -> "CLM-BAS-F2", "RootMIDI" -> 41, "BoreID" -> 0.875 | >
};

holeOffsets = {1, 2, 4, 5, 7, 9, 10, 11, 12};
holeRatios = {0.32, 0.36, 0.38, 0.34, 0.38, 0.40, 0.30, 0.34, 0.32};

familyTable = Table[
With[{v = variants[[i]]},
< |
"ID" -> v["ID"],
"RootHz" -> midiFrequency[v["RootMIDI"]],
"BodyLengthIn" -> bodyLength[v["RootMIDI"], v["BoreID"]],
"HolePositionsFromBellIn" -> MapThread[
holeXFromBell[v["RootMIDI"], #1, v["BoreID"], #2]&,
{holeOffsets, holeRatios}
]
| >
],
{i, Length[variants]}
];

Dataset[familyTable]

Manipulate[
Plot[
speedOfSoundInPerSec/(4(x + 0.61(bore/2) + 0.25bore)),
{x, 4, 42},
PlotRange -> {80, 700},
```

```
AxesLabel -> {"Body length (in)", "Frequency (Hz)"},
PlotLabel -> "Stopped cylindrical reed bore first-pass model"
],
{{bore, 0.5, "Bore ID (in)"}, 0.35, 1.0}
]
```

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README.MD

Project artifact.

CHALUMEAU FAMILY

> A build-ready design packet for a family of single-reed chalumeaux:
> keyless folk-pipe simplicity first, optional handmade two-key metalwork
> second, and a documented path toward clarinet-style register experiments.

WHAT THIS IS

This repository now contains a complete first-pass engineering packet for designing and building a family of chalumeaux in soprano C, alto G, tenor C, and bass F. The design is parametric: body length, bore, tone-hole positions, and optional levers are driven from formulas rather than hidden one-off dimensions.

The packet starts from the local inspiration photos in images/, Tony's Musical Instruments.xlsx workbook, and especially the reed/bore lessons in the Great Highland Bagpipe sheet. The acoustic model is not copied from Native American style flute K2 corrections. A chalumeau is a cylindrical, single-reed, effectively stopped pipe, so it gets its own validation loop.

FAMILY PLAN

ID	Variant	Root	Bore ID	Final body L	Blank L	Sections
CLM-SOP-C4	Soprano C	C4	0.5	12.672	14.172	1 body + mouthpiece + bell
CLM-ALT-G3	Alto G	G3	0.625	16.939	18.439	1 body + mouthpiece + bell
CLM-TEN-C3	Tenor C	C3	0.75	25.483	26.983	2 body joints + mouthpiece + bell
CLM-BAS-F2	Bass F	F2	0.875	38.32	39.82	3 body joints + mouthpiece + bell

The recommended build order is CLM-SOP-C4 first, with no levers, using a commercial reed/mouthpiece. Once that speaks cleanly, add the two optional levers to the same body or a second soprano body. Then scale up to alto, tenor, and bass.

WHY SOME HAVE LEVERS

Keyless chalumeaux are mechanically close to folk reed pipes: finger holes cover the notes that are reachable by hand. A few levers appear when a useful tone hole is too low, too high, too large, or too awkward to cover directly. The historical two-key chalumeau/early clarinet moment is exactly that transition: keys first helped extend and smooth the low register, then the clarinet moved one key into a better register vent position and gradually accumulated more keys for chromatic notes, trills, low extensions, and better intonation.

PACKET MAP

- chalumeau-family-design-table.xlsx: Excel design workbook with blue inputs and formulas.
- family-spec.csv: one row per family member.
- data/tone-hole-schedule.csv: calculated tone-hole and keyed-hole positions.
- design.md: governing model, assumptions, keywork strategy, and risk register.
- hardware/lever-fabrication-guide.md: how to make the metal levers, pads, pivots, and springs yourself.
- cad/solidworks-design-table.csv: SolidWorks configuration table.
- cad/solidworks-global-variables.md: variable naming conventions and equation pattern.
- drawings/: SVG drawing sheets for the family, soprano C, and keywork.
- assembly-manual.md, bom.csv, sourcing.csv, cut-list.csv, validation.csv, supplier-rfq.md: shop-facing build packet files.
- wolfram-starter.wl: Wolfram starter for acoustic sweeps and tuning validation.
- capstone-deck.pptx and print-packet.pdf: presentation and printable versions of the packet.

STATUS

Area	Status
Acoustic model	first-order stopped cylindrical reed model complete
Parametric family table	complete, with formulas and SolidWorks export
Keywork concept	two-key handmade lever system specified
Manufacturing drawings	SVG first-pass sheets complete
Shop build method	complete for prototype workflow

Tuning validation	template ready; needs measured prototype data
SolidWorks	variable conventions and design table ready

LICENSE

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SKILLS.MD

Project artifact.

SKILLS DEMONSTRATED

STOPPED-CYLINDRICAL-REED-BORE

Models a single-reed cylindrical bore as a stopped pipe in the low register, with explicit reed and bell end-correction assumptions.

HANDMADE-WOODWIND-KEYWORK

Designs and fabricates simple early-woodwind levers using brass or nickel silver, pivot rods, pads, posts, and springs.

SOLIDWORKS-PARAMETRIC-FAMILY-TABLE

Uses one design table to drive multiple instrument-family configurations, with stable named dimensions that match CSV, drawings, and validation logs.

EMPIRICAL-TONE-HOLE-VALIDATION

Starts with physics-derived hole positions, then tunes the real prototype by measured frequency and cents error rather than trusting the first-pass model blindly.

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FAMILY-SPEC.CSV

Project artifact.

instrument_id	variant	root_note	root_midi	root_frequency_hz	bore_id_in	wall_thickness_in	body_od_in	bell_od_in	acoustic_length_in	body_length_final_in	blank_length_in	turning_blank_in	sections	build_priority	status

CLM-SOP-C4	Soprano C	C4	60	261.63	0.5	0.188	0.875	1.65	12.95	12.672	14.172	1.25 x 1.25 x 14.172	1 body + mouthpiece + bell	Prototype 1	first-order design; validate after reed and bore prototype
CLM-ALT-G3	Alto G	G3	55	196.0	0.625	0.219	1.063	1.95	17.286	16.939	18.439	1.438 x 1.438 x 18.439	1 body + mouthpiece + bell	Prototype 2	first-order design; validate after reed and bore prototype
CLM-TEN-C3	Tenor C	C3	48	130.81	0.75	0.25	1.25	2.35	25.9	25.483	26.983	1.625 x 1.625 x 26.983	2 body joints + mouthpiece + bell	Prototype 3	first-order design; validate after reed and bore prototype
CLM-BAS-F2	Bass F	F2	41	87.31	0.875	0.313	1.5	2.9	38.806	38.32	39.82	1.875 x 1.875 x 39.82	3 body joints + mouthpiece + bell	Stretch prototype	first-order design; validate after reed and bore prototype